

Getting Entangled with Sean Hodgman

Podcast: Startalk Podcast

AI Executive Summary

Core Thesis The episode explores the profound complexities and recent advancements in quantum entanglement, specifically focusing on a groundbreaking experiment by physicist Sean Hodgman. While entanglement has historically been demonstrated using massless photons (via polarization) or stationary atoms (via spin), this research demonstrates momentum entanglement in massive helium atoms within a Bose-Einstein Condensate (BEC). By proving that the physical path/momentum of atoms can be entangled, the research opens new avenues for studying how gravity interacts with quantum mechanics and advances the potential for large-scale quantum computing and unhackable communication.

Key Arguments & Data

- * **Bose-Einstein Condensates (BEC) as a Tool:** By cooling helium atoms to approximately one-millionth of a degree above absolute zero, scientists can eliminate random thermal motion, forcing atoms into a single, coherent quantum state (a "blob") where they behave like waves rather than individual particles.
- * **Momentum Entanglement vs. Previous Methods:** Traditional entanglement often focuses on **spin** (up/down) or **photon polarization** (vertical/horizontal). Hodgman's experiment achieved entanglement in **momentum**, meaning the atoms exist in a superposition of multiple travel paths (e.g., northeast/southwest and northwest/southeast) simultaneously until measured.
- * **The Einstein-Bell Debate:** Albert Einstein famously dismissed entanglement as "spooky action at a distance," arguing that quantum mechanics must be incomplete because entanglement appeared to violate his principle that nothing travels faster than light.
- * John Bell's **Inequality Theorem** provided the mathematical framework to test this; subsequent experiments (including those by Alain Aspect) proved that quantum superposition is real and violates classical physics predictions.
- * **The No-Communication Theorem:** A critical distinction is made regarding "faster-than-light" communication. While measurement of one particle instantaneously affects another, entanglement **cannot** be used for faster-than-light messaging because the outcome of the measurement is random; a "classical" signal (limited by the speed of light) is still required to interpret the results.
- * **Quantum Computing and Cryptography:**
- * **Computing:** The transition

from bits (0 or 1) to qubits (superposition of 0 and 1) allows for massive parallel processing in specific problem sets. * **Security:** Entanglement enables "provably secure" encryption; any attempt by an eavesdropper to intercept a quantum message would disturb the state, alerting the communicators to the breach. * **Decoherence and Scaling Challenges:** Maintaining entanglement is extremely difficult due to "decoherence"—interference from external factors like stray magnetic fields or heat. While scientists have entangled groups of up to a thousand atoms, scaling this to larger molecules or macroscopic objects remains an active area of research.

Actionable Takeaways * **Monitor Quantum Computing Volatility:** For those interested in the intersection of finance and technology, keep a close watch on "quantum arms races." The shift from theoretical physics to engineering-based quantum computing represents a significant long-term economic and security inflection point. * **Understand the Limits of Information Theory:** Use the "No-Communication Theorem" as a mental model for understanding data limits; even with highly connected systems, the fundamental speed of light remains a bottleneck for information transfer. * **Support Fundamental Research:** The episode highlights that major technological revolutions (like the IT revolution) stem from "curiosity-driven" basic research into phenomena that initially seem to have no practical use. Investing in or supporting scientific literacy and basic physics funding is essential for future technological sovereignty.

Full Transcript

Chuck loved me some quantum physics, and apparently so does everybody else, especially when we're talking about quantum entanglement. I'll say some of the best questions we've received on quantum entanglement, yeah. One of the world's experts we went down under to Canberra, Australia, Sean Hodgman coming right up, a physicist on top of the situation. Welcome to Star Talk, your place in the universe where science and pop culture collide. Star Talk begins right now. This is Star Talk, Cosmic Queries Edition. What is the subject? Quantum entanglement. Chuck, are you ready for this? Never. I mean, I'm just going to be honest. Haven't you and I been quantum entangled for a while? Yes, without a doubt. Any time you feel pain, Neil, I feel it.

Immediately. Oh, here you go. That's how that works. You see? You see? Now, I have like a storybook understanding of quantum entanglement. So to really get to the bottom of this, we combed the world to find somebody who actually works in the field. And we found a physicist at the Australian National University, A&U at Research School of Physics there. And that would be Sean Hodgman. Sean, welcome to Star Talk. Thanks a lot for having me. Yeah, this is everybody. Nobody doesn't like quantum entanglement. Everybody's into it. And we have a million questions. Before we even get to the queries part of this episode, we have questions of our own. I have questions of my own. And so let's just come right out of the box and tell me what you publish papers on.

What is it you do? Yeah, so my group works on a whole range of experiments. Our particular apparatus that we work on involves making helium atoms really cold. So we take them and we cool them down to almost absolute zero. So absolute zero is as cold as you can get when there's essentially no motion in the system. Again, remembering that thermal temperature is basically just random thermal motion. We take all the motion out of the system and we make it really cold the temperatures we get to are a millionth of a degree above absolute zero I'd say that's cold Listen man. Yeah, you're almost there. Okay Keep at it man. Keep at it. You're almost there man helium it was familiar as a gas you can inhale out of a balloon in

a in a birthday party. Remind me, it liquefies around three degrees, is that correct? So normally, at normal pressure it would, but because we do it in a vacuum system, we keep it in the gaseous phase. So it's still a gas at these really cold temperatures, just at really low density. Whoa, cool. Okay. And so why the hell do you do that? Yes.

We ask ourselves that question too sometimes. So the purpose of this, at these cold temperatures, all the atoms will form a single coherent quantum state called a Bose-Einstein condensate. So that's where quantum mechanically low temperatures, so really cold and when atoms are moving really slowly, they don't behave like these little billiard ball situations that we like to think of. What they actually behave like is they become these fuzzy, smeared-out quantum blobs. And at these temperatures, they all become essentially an identical quantum state, which is very similar to a laser, where a laser is the same for photons. Is this the same thing as you cool it down? It's effective wavelength increases? Exactly, yeah. Is that a fair way to say that? And so that the wavelength is so long, they're all just sharing the same wavelength. And so they have a high of mind at that point.

Is that a fair way to characterize this? Definitely. So the reason we don't see these quantum effects of particles behaving like waves at normal temperatures is because their wavelength is too small to actually see. Once you get to the temperatures that we cool them down to, then they form the single quantum state and their wavelengths are actually, it's macroscopic. So it's inside the trap, it's maybe 100 micrometers, so 0.1 of a millimeter. I'm not great on inches, so for your American use, I'm very short. Well, it's anyone's over, if you don't know inches. And when we drop the atoms from the trap and release them onto our detector, so they fall nearly a meter, and in that distance, they expand, and so by the time they hit the detector, they're actually sort of centimeters big. And so we have a quadruple object that is on the centimeter scale by the time it hits the detector. That's amazing. And when you say blob, what kind of structure, because in a gas, it's almost random. They're just sliding around everywhere. But in a structure, sometimes it's like a lattice. So what does the quantum state look like? Yeah, how do they look like compared to each other? Yeah. So because they're all identical, they're essentially an identical particle. So it's basically just one state that just looks a very smooth sort of blob essentially.

There's no random motion, so they're not really bouncing around off each other. There's just essentially one smooth blob. I remember first reading about this many years ago and was just totally impressed. And it's got Einstein's name on it. And Bose was an Indian physicist, right? And we credit him for the word we use for bosons, the particles. Is that right? Yeah, exactly. So the particles we call down are bosons. We can also call fermions, and they do a whole completely different set of physics, but yeah, the two types of elementary particles are bosons and fermions, and for boson-Einstein condensates we use bosons, which, yeah, were named after the Indian physicist Bose, who came up with the statistics to describe them, and to describe this state of boson-Einstein condensation. No, Einstein was not a fan of quantum physics, so why did he get something named after him? What's up with that? Yeah, I mean, it's kind of funny that Einstein. He essentially was one of the inventors of quantum physics. It was his black body radiation paper that came up with a lot of the initial physics stuff. What he had a problem with was some of the aspects of physics and kind of the interpretations of quantum physics were what he really struggled with a little bit, some of the things such as entanglement. So he didn't like the fact that when you have an entangled system, if you have two particles, it essentially means that if you take two particles and you separate them, if you measure one of them, you'll instantaneously know the state of the other. And this is what Einstein didn't like because that implies that something travels faster than the speed of light, which famously violates one of his other famous works on relativity. And so he really didn't like that. And yeah, so he worked hard to sort of say that, well, quantum mechanics must be incomplete. There must be a way that there must be some information that we're missing here.

And it was only sort of maybe 30, 40, 50 years after that that we were actually able to prove that, no, that really is how the world seems to work. Yeah, so Einstein was wrong in his assumption that it's incomplete, I guess. Or maybe it's still incomplete philosophically, but everything works. So no, right? He was just upset because it made him look, it made his suppositions look stupid. That's what was wrong. He was like, don't you know I'm Einstein? Do you know how smart I am? This can't be the case because now everything that I figured out in my physical representation of the universe

can't be. So guess what? No, no. The Chuck account of the history there. So Sean, where does entanglement come in to this Bose-Einstein condensate?

The Bose-Architect condensate, we essentially just use as a source for our experiments on entanglement. So what we do is we take our condensate, we split it in two, and we collide the two parts of the condensate together. So we basically give one half of it a kick with a laser beam, and then it collides with the other half. And in those collisions, you get all these individual pairs of atoms from each of them will collide off each other, and they could go in all sorts of different directions. Let's just focus on two of the directions they could go. So if you have two atoms, I can probably do it with, I normally do this with a couple of coins. So if you have a couple of, imagine these coins are the atoms, you bounce off each other, they can either go that way or they could go that way. Okay? So either this way or that way. And classically, if we think of them as little billiard balls, they could either go this sort of up or down. It doesn't, they do either of those. And when you do the experiment, they go this way, you measure it.

or they go this way and you measure it. Now, quantum mechanically, that's not what happens. What happens quantum mechanically is that when they do the collision, they go this way and they go this way at the same time. For those of you who might be listening without the benefit of YouTube, this way and this way is kind of northeast and northwest instead of northeast and southwest. And that's what you're saying. quantum mechanically both of those pairs Well, the atoms go both ways However, when you measure it, there's only still two atoms in the system. We're not we're not creating matter here We're not creating anything and so when you measure it, you'll only either get northwest southeast or northeast southwest But until you measure it the atoms have gone both paths and that's that's what entanglement oh my god, okay, so now I've only ever heard of entanglement with regard to particles. And now you're describing an entire atom. At those temperatures, the hydrogen nucleus will have its complement of electrons. So you've got a whole fricking atom here. And are you saying you're entangling atoms? Wow. Yes, exactly. So most entanglement so far has been done with things like photons, which is a very elementary particle. It's just an excitation of light.

But like you said, Neil, I mean, Helium atom is quite complicated. It's got a nucleus, which has two protons, two neutrons. It's got two electrons whizzing around that. And now we're taking two of these and we're entangling them together. First of all, that's crazy. Based on everything that, you know, so far. It's time to see where Einstein was coming from. I am. Oh, you're a good company, Chuck. OK. Yeah. Damn. I mean, that's kind of insane. You know, so now All right How long how long do they stay entangled because that's a big Contest out there, right? I mean I remembered reading some papers about what they were doing in China where they had two entangled particles one was in orbit and what was in a lab and so It's like how they doing this and what are they after is a distance or is it time? Or do you have to make sure nobody messes with one of the particles so you have to You have to make sure it stays isolated. What are the conditions to sustain this? Yeah, that's pretty much all correct. So in our case, we're entangling our particles in momentum or in the path that they travel, so the direction they go. And that's really hard to keep your particles entangled, because if they go slightly differently from that path, so instead of going northeast, southwest, they go slightly towards north northeast, say, then you're no longer going to be entangled because you've now gone on a different path. You've broken the symmetry of it, I guess. Yeah, you might collapse the entanglement or degrade it to an extent. So our particles don't stay entangled for long. Our whole experiment's about a millisecond.

that we do this entanglement for? A millisecond is not a millionth of a second, even though it sounds like it should be. A millisecond would be a thousandth of a second, correct? A thousandth of a second, exactly, yes. Right, like a millimetre is a thousandth of a metre. So a millisecond is a thousandth of a second. Okay, so these are not very long. I mean, that's an eternity on a quantum scale. It all depends on your perspective. Right. Yeah. Is there a long-term goal for this experiment or is it sort of an existence proof that you can entangle atoms? Yes. So this particular experiment that we did, it was kind of a demonstration that we could do this. So people have, there's been a lot of entanglement previously done with photons. There's been some entanglement done with atoms as well. But the entanglement that's been done with atoms hasn't involved external degrees of freedom.

by external degrees of freedom, I mean basically the fact that it moves in different paths. So things like momentum. Previous experiments have just been things such as spin. So you might put it in a superposition of being in different states, but they stay at the same place. And so ours was the first experiment that showed momentum entanglement with atoms. And the reason why that's interesting is because One of the things you might want to look at atoms for over photons is that atoms interact much more strongly with the gravitational field. And so, potentially, we could look at effects such as how does gravity interact with entanglement down the track we're talking about. And so, that might open up avenues to explore things such as quantum gravity theories. It's not that gravity doesn't interact with the photons, it would just be much harder to measure, right? So, whereas a tangible particle You've got something that's path you can track, I guess. Is that what's going on there? Yeah, exactly. It's a much stronger interaction. I'm Joel Cherico and I support Star Talk on Patreon. This is Star Talk with Neil deGrasse Tyson. Hey, Sean, can you help me out as I'm trying to wrap my head around this? With the undetermined state and the path, because you said If it goes on a different path, can you talk a little bit more about that? Because I'm not quite understanding the superposition before the actual measurement. Because if China has a particle in orbit and a particle in the lab, that sounds like they're on different paths. So where does the sensitivities come from for changing what one particle does relative to the other?

Well, thank you, Neil. That was my question, Sean. In our case, it's the fact that it's the momentum states of the atom. So it's the direction it's traveling that is the entanglement. So if you imagine sort of north, south, sorry, northeast, southwest, northwest, southeast pairs, then you can imagine that It's the fact that the atoms are either going northwest, southeast, or they're going northeast, southwest, and they're going those two directions at the same time. Now, what's happened with previous atoms is you'll have an atom sitting somewhere, and then you'll have another atom sitting somewhere, and you'll use photons to communicate between those two and to flip the atom into a particular state. So it's still sitting there. It's doing whatever it's doing. Yes, it might be orbiting Earth. It might be sitting in a lab, which is, of course, rotating this Earth.

So it is moving, but it's not the motion that's entangled. That motion doesn't change. So the in momentum is really the key here. I got it. OK. I got it. And the photon is your measurement of the state, right? Because you can't measure it unless you interact with it in some way. For our atoms, it's actually slightly different. So. No, I meant for the other case. In the other case. OK. Got it. Got it. Got it. OK. All right. So I have my notes here. to inquire with you about the Bell inequality theorem? How's that relevant to what's going on here? Yeah, so historically, if we're going back to Einstein and what he didn't like about entanglement, so he didn't like the idea that you could be in these two states at once and then measure, and that would collapse your superposition so that you'd then know which way to go. Was that or was that not his invocation of the phrase spooky action at a distance?

Yeah, that was exactly what he said. So he described it as spooky action in existence and he says, you can't have this, you can't have this, that you'll collapse to being in this state. But before you collapse it, you're in both states once. He didn't like that. He thought there must be something. Am I correct in that? Because I've heard two different answers, but you're the horse's mouth here, that the other particle knows of this not simply faster than light, but instantaneously. Which of those is the right way to think about it? To the best of our knowledge, it seems to be instantaneous, but it's very hard to prove that it's exactly instantaneous. Wow. People have proven that it's faster than speed of light, though. So tell me about the inequality theorem. Yeah. So, Einstein and his co-authors, Badovsky and Rosen wrote this paper, which was the famous Spooki aficionado distance paper.

saying that this can't be how the world works and there must be something in quantum mechanics that's incomplete. And everyone for decades kind of thought, well, you're never going to be able to test this, so it's just a philosophical debate to an extent. And then fast forward sort of 30 years or so into the 60s, and John Bell, there was another famous theorist, came up with an experiment where you could actually measure this. Because the problem is with entanglement, if you imagine that you're in this superposition of northeast, southwest, and northwest, southeast, but you only ever get one result out of the system. So it's very hard to prove the difference between being in that superposition and not being in that superposition. And so what John Bell said is, hang on, we'll take that

superposition and we'll then interfere it back with itself. So if you imagine if you have the particles going northeast, southwest, northwest, southeast, you then take those two parts of the superposition and you reflect them back on each other so that you have particles go like this and particles go like that, and you end up, there's a spot where they overlap again. And because the halves of the superposition can overlap, you can then get quantum interference at that point. And so Bell came up with this inequality called the Bell inequality, which he said that if they really are in this superposition, you'll get this interference. And he As far as we know, thought that his inequality would always hold, that classical physics would be correct, and that the quantum prediction, which predicts that his inequality is violated, wouldn't be correct. And then, yeah, a couple of decades later, some physicists, such as Alain Espay, and co-measured it and showed that it is actually.

So that was a thought experiment, not an actual experiment that he conducted. Wow. Right. Right. Let's go to our fan base, our Patreon supporters, each paying 5 a month to gain access to our guests in the form of a question. Yes. Let's start with John Mayer. And John Mayer says, Dear doctors, Tyson Hodgman and Lord Nice, I've been reading about entanglement for decades, but I have never felt I could understand its nature. So thank you for this episode. My question is three parts. So one, how do you entangle a particle? And two, how do we know they are entangled with spooky etching and not just similarly related? So I love the show and bravo. And let me jump in the middle and ask, can you just take two random particles and forcibly entangle them?

or must they be birthed together to be entangled in the way you describe? So you need to... I'll go to Neil's point first. So you just need some way that those particles can be identical. So in our case, we used identical helium atoms, but you wouldn't need to. For instance, if you had non-identical particles, you could just collide them off each other, and one would go one way, the other would go the other way, and they'd be in a superposition of say let's let's again we'll go back to my coins say we have sort of a red and a blue particle you could collide so red goes one way blue goes the other way or blue goes one way red goes the other way um that we could find as well a red and a blue particle yeah yeah i did not know that i thought they had to

be kind of sort of symmetrically identical you know with just complementary uh of elements like spin or whatever. So that's interesting to me. Okay. So in practice, how are you entangling particles? That's the first question, right, Chuck? Yeah. Yep. Yep. So in practice, we take one particle and we give it a kick and we collide it with the other particle and they bounce off each other. And then you get in this superposition of the particles going as we're talking about northeast, southwest or northwest, southeast.

So you're in these different momentum states, and that's your initial entangled state. So the act of colliding them off each other, brings their wave functions into harmony. So what was two wave functions becomes one? Yeah, I think that's a really good way to describe it. There's two separate wave functions that you can describe separately. And then after you collide them, there's no way that you can describe them with two different wave functions. You have to use a single wave function. And this is because the wave particle duality of nature at those smaller scales. This is quantum physics at its finest, right? So the second question was what, Chuck? actually entangled with spooky action and not just similarly related. That seems to be the real question there. Yeah. Yeah. So I think that that kind of goes back to, again, if they were classical particles, you could put them in this, you could put them in kind of a superposition where they either go one way or the other way. But quantum mechanically, they go one way and the other way at the same time. And so the reason we can and the way we can prove that even though we only ever measure one outcome, is that if we then take the two halves of that superposition and we combine them back together and interfere them, then we can show that there will be different outcomes. And so the particle can essentially interfere with itself. And we can show we'll get different outcomes to what you would get classically. Wow, man. That's all right.

That is some freaky stuff. I love it. It's freaky. It's freaky Friday. Yeah. So this is Hayden Gorin, I think. Gorin. He says, hey, Dr. Tyson, Dr. Hodgman, Lord Nice, hope you are all doing well. It's in the paper by Dr. Hodgman at all that the helium atoms were momentum entangled. And I hadn't heard of sub types of entanglement before. What? Other types of entanglement are there. And why did the team opt to use momentum

entanglement for the BEC made of helium atoms? All the best Hayden from London, England. What is the inventory of entanglements that you have? Yes, there's lots of different types of entanglement. So if you have the original experiments with photons, it would often be something such as polarization.

So polarization is just, if you imagine a photon as a small particle of light, it's basically which way is the light vibrating? Is it vibrating this way or is it vibrating this way? So vertical or horizontal? And you might entangle your photons in that so that one of your photons has vertical polarization, one has horizontal, and so you could be in a superposition of vertical going right, horizontal going left, or horizontal going left, vertical going right, and then you could measure that. And yeah, and so that was a lot of the original experiments were done with photons with things such as polarization.

Atoms, previous experiments have done things such as spin. So spin is just a fundamental property of atoms or charged particles that you can have sort of spin up or spin down. And you could be in a superposition of spin up and spin down and entangled with that. And in our case, we did momentum. Pretty much any quantum property can be entangled.

You just have to be able to put it into a superposition of those states. Well, let's get back to the spin. So in the two entangled spin particles, does one have to be spin up and the other has to be spin down? No, no, definitely not. As physicists, we tend to use a shorthand that if there's any system that is a two spin system, you'll call that spin up and spin down because that's how we tend to learn about it. undergraduate physics and we tend to just Yeah, yeah. So for instance in my lab we've done previous experiments where we've used a spin one and a spin zero state of helium and we've entangled them and But we still call it spin up and spin down because it's just it's shorthand that physicists understand easily because when I think of quantum products I think they have complementary Quantum states, but that's not the case they can have any quantum state You just have to be able to couple between those quantum states. If you can't couple between the quantum states, then you can't interfere them. So if you can't change the, coupling just means changing. So if I can't change controllably from spin one to spin zero or spin up to spin down or horizontal to vertical polarization, whatever your entanglement parameters are in our experiment,

it's momentum. And so as long as you can change coherently between those states, you can get entanglement and show that you've, show of belly inequality violation. Okay. Wow. Look at that. That is mind-bending stuff. I did not know that. I did not know that. Okay. Bring on some more. He says, hello, Dr. Tyson, Dr. Hosman. This is Jonas Williams from New York City. My question for you.

Has the study of quantum entanglement and subatomic particles changed or influenced how you understand reality? I like that. I like that. If you're not purposely entangling particles, is that something that can happen to them by natural causes? Oh, definitely. Yeah. Yeah. Entanglement can happen all the time. It's just that it's because it happens on such a small scale. We normally don't see the effects of it. And so, yeah. And I think that, back to the question, for me, it's really weird. And I think a lot of us who have studied quantum mechanics when you first encounter it, whether it's in an undergrad or high school or through a podcast like this, when you first hear about these quantum features, you might think, yeah, that sounds really weird. That can't be how the world works.

Because it's not how our brains have evolved. Our brains have evolved to be used to throwing things at animals, I guess, which is a very classic way of looking at the world. And if I throw a cricket ball, what do you have? Baseball in America? If I throw a baseball, let's say. And it's not going to be in two places at once. It's not going to go two directions at the same time. And so, when we read about quantum mechanically that a small particle, which we like to think of as just a smaller ball, when we read that can go both directions at once, we just sort of think, well, it must be some mathematical trick. And the fact that we can actually do these experiments to show, no, no, at the very small scale or very cold, this is how the universe works. It's a bit mind-bending, right? Yeah, without a doubt. To summarize, because it happens on these microscopic scales, there's no macroscopic manifestation of entanglement that we experience. Because we're just trying to not get eaten by the lion, and that doesn't require quantum physics to accomplish. Are they forever separated, the quantum and the macro physical? Or is there a possibility that there is a point of crossover?

Yeah, that's that's a really good question and that's that's a really active area of research at the moment that Because clearly at the small-scale quantum works at the large-scale quantum doesn't But there's got to be somewhere in between There's either must be a hard point where they stop working or maybe it's a fuzzy boundary a lot of the research We're kind of seeing these days isn't there might be a fuzzy boundary where you just kind of get less and less quantum and more and more classical Wow, all right God, how do you deal with this every day, man? All right, here we go. Do you arrive at home at the end of the day depressed or jubilant? Normally depends on how much university bureaucracy I've had to deal with in a particular day. The quantum physics is the easy part of the job some days.

As they say, the administration is a new kind of subatomic particle called the moron. Damn. Okay. This is Mike Parker. And Mike Parker says, Dr. Hodgman, Dr. Tyson, Lord Nice, Mike Parker from Virginia. It seems entanglement is observed in larger and larger particles. Is there an upper size limit to entanglement? Could there ever be a way to use particle entanglement for instant long distance communication or other practical purposes? This is the question that every sci-fi fan wants to know, man. Can we have faster than light communication like on Star Trek where captain, there's a subspace communication waiting for you in your quarters. Or can you entangle molecules? I mean, yeah, we're impressed with the atoms, but what have you done with us lately? Can you can you entangle molecules be they simple or complex? On the entangling larger particles, that's that's another really active area of research people keep pushing it further and further that there have been people that have entangled collections of atoms and molecules and they sort of push out I believe I believe it's of order a thousand atoms, but don't quote me on that because I could be I could be wrong on that It's all I care about. Yeah, it's more than way more than one. Yep Yeah. And so, yeah, trying to push this entanglement on the bigger scale and work out where it breaks down. At some level, it clearly doesn't, it doesn't work anymore. And so, yeah, where that transition is a really active area of research. So maybe you were familiar with George Gamma's Mr. Tompkins and Wonderland. I don't know if I'm a little older than you, so I don't know if this was around in your day. But George Gamma, Mr. Wonderland was a place where the laws of physics were different.

simply by the physical constants being different. So in one of them, the speed of light is like 60 miles per hour. And so you're driving down the street and then you see these relativistic effects just by approaching the speed of light as your speed limit on the road. One of them was they changed the value of the Planck's constant so that macroscopic objects would feel Quantum phenomena that you'd walk through the door and you'd like diffract. It's just it's fun to think about what how that if you have a knob that could tune it What different things would happen to you or we just simply develop a new sense of? What the world is right like you said we throw the ball and it follows one arc but if quantum rules were all around us at all times, and you threw the ball and it split into two states, you would say, oh, it's just splitting it into two states. It wouldn't even be odd to watch that because it would happen so frequently. Is that a fair way to think about this? Yeah, I think so. Our brains are great at adapting, and I'm pretty sure that if we'd had to evolve to live in a quantum world, I'm sure we'd just think it was normal. It was just normal. That's right. That's cool. And what about the communication?

a portion of Mike's question. Would it ever be a viable use for long distance communication? Is there a future in this? Yeah, so this is what everyone thinks as soon as you hear entanglement. You think there's instant, measuring one instantly changes the other. We could use that to communicate faster than the speed of light. Unfortunately, that doesn't seem to be the case. And essentially, it's because the way you measure it contains information. If you make a measurement on that particle, yes, you will know what the other one is, but you won't be able to communicate that in a useful way unless you communicate it classically because the other person would then have to make a particular measurement to get anything useful out of that. You can't actually use it for faster than light communication. There's a theorem called the no communication theorem, which basically says that you can't on.

You can't use entangled states to communicate faster than the speed of light. Really? That's the name of the theorem, the no communication theorem. That's the best name they can come up with. I believe that's what many women call their husbands. The no communication theorem. So basically what you're saying is the collapse of the superposition and once that

information is set, you can't know it unless you're making the same measurement that the other person is making on the other side. And you wouldn't be able to know which position it is unless you were to call them and say, here's the position. Yeah, it's exactly because quantum measurement perturbs the state. If you make your measurement in a way that could exploit that information, it will change the state of it.

And so then that will change the state of the other one. And then when they make their measurement, they'll get a different result out of it. They'll get a different result. Unless you've called ahead and told them how you need to make this particular measurement. Which, by the way, is how you know that it's entangled. That's how you know it's entangled. Exactly. It's kind of frustrating. You're like, we should be able to do this, but yeah, we can't. So we can't use it for encrypted message sending. Now, that's a completely different question. You can use entanglement for encrypted message sending. That's where you exploit the fact that if you measure one half of that, you will change the other half. If you encrypt your message on two photon pairs and you send one of them to the person that you want to and you keep the other one yourself, then if you make measurements on that and compare the results, with the person you've sent it to, you'll know if anyone's messed with your system because you'll get different results. And so you'll know if it has been eavesdropping on your messaging. And so there's mathematically provably secure encryption protocols using quantum communication, which can be shown that you just can't break them because if you interfered with it, if you interfered with the state, if you listened to it, you interfered with the state and people would know your eavesdropping and you could just abort the communication.

So this is a pipe dream then, it's a pipe dream that people have for it. These things have been demonstrated at various scales and yeah, they're essentially, it's almost an engineering problem at this stage to get it to work better. Okay, look at that. All right, this is William Warren and William says, hi, Dr. Tyson and Dr. Hodgman, this is William Warren from Abingdon, Maryland. If quantum entanglement is a fundamental feature of nature, could space time itself emerge from a vast network of entangled particles? In other words, is it possible that distance isn't fundamental, but rather a consequence of how information is connected at the quantum level?

Thank you so much. We've heard on another installment of StarTalk from one of our physicist friends, Brian Greene, that it might be that the virtual particles in the vacuum of space that are connected to each other by entanglement, they're entangled, that that entangled gap between them may be a wormhole. And a wormhole would have the same property because you would just step through And you're just there, right? You're not moving faster than light. The whole enabled that. And it's not because you had special rockets. So what is the latest thinking other than what I just shared with you about what the entangled pathway actually is?

Yeah, wow, this is um, you don't remember. I'm only a dumb experimentalist. Um, I basically I would certainly defer Um, yeah thinking on that to things such as to the experts like Brian and um and code cuz um, yeah That's that's kind of a bit a bit beyond. Um, what we're working on. I it certainly sounds like a um uh interesting take uh but i'm yeah i'm not not really sure unfortunately sorry okay no okay but listen we we like that answer uh you know now is this hard to get somebody to say hey i'm not that sure sorry you know what i mean yeah it's a really good point i think it's a big part of being a scientist is you've got to learn to know what don't you know and it's one of those things that the more you know the less you know, you know, it's kind of the reverse Dunning-Kruger effect, you know, the Dunning-Kruger effect, but the less you know, the more you think you know, it's kind of the reverse. The more you know, the more you realize that, wow, there's heaps of stuff that I just don't know at all. And, yeah. Dunning-Kruger, we get along very well. Very well. Dunning and Kruger, I know them. We have a great relationship. They told me I do the best Dunning and the best Kruger. All right.

I think there's I think there's a value in saying I don't know and allowing people to understand that that science is sometimes the answer is, well, we don't know. You know, absolutely. He's not knowing. They're too different. The science doesn't know. And I don't know those two different things. Right. And they're both the same for me. Whether physics itself knows. I think the answer is still out on that as well. I think there's still some open theories in that, because again, you can come up with a theory, but you have to be able to prove it experimentally. And I think to prove something like that experimentally, it would be extremely hard. So there's there's some

really interesting theories, and the trick is going to be a bit like with Einstein's spooky action in a distance claim, coming up with an experiment to test it. That's that's what will be. Yeah, that would be a big Nobel Prize discovery. Cool.

All right, let's go to Alejandro Guardado. And he says... He's from Hackensack, New Jersey. No! Where's he from? Where's he from? I don't think Alejandro says where he's from. Oh, okay. He says... He's from Monterey. No, Washington State. I'm sorry, he did say Washington State. Washington State speaks that way. I just want you to know.

That's my fantasy of Alejandro in Washington state. He says, hey, Dr. Tyson, Dr. Hodgman, nice Alejandro here from Washington state. My question is, what stabilizes quantum fluctuations when particles fuse or collide? How does this increasing proximity affect entanglement more extremely? How would this work for particles in the singularity of a black hole? Thank you. Yeah, so I want to reshape that just a little bit. So if I have two particles, we know that quantum physics says there's everything is always in motion. There's always some energy to the state. Does that disrupt any attempt to entangle two particles? So in other words, can two particles natively break apart simply because of quantum fluctuations that are inherent in all particles and all systems. Quantum fluctuations at some level will probably have an effect on entanglement, but it's normally at such a small scale that it won't stop entanglement happening, I think. In fact, when you cool down your helium atoms, you are reducing the quantum fluctuations by dropping the temperature, correct?

We're kind of reducing the classical fluctuations. So temperature is really a classical phenomenon. It's just random motion of particles. We're reducing the classical fluctuations to get to the scale where you can, in principle, see quantum fluctuations. However, for our entanglement, for our particular system, the quantum fluctuations don't really come into it. There's much larger things that cause the entanglement to deco here, things such as our classical, so magnetic, stray magnetic fields and the like. Your microscopic disruptions to it. Yeah, definitely. We're not really at a level where we're sensitive to the microscopic quantum fluctuations. This is Mikael Boisvert

who says, hello, Guardians of the Geeks. Mikael here from Canada. I like that, Guardians of the Geeks.

How would the universe change if the programmer behind would suddenly toggle particle entanglement off? In the same spirit of that, I heard someone suggest that evidence we're in a simulation is that the programmer already put a limit to how fast things can go, because they can't simulate it faster than that. So the speed of light is the programmer's limit that we've bumped up against. And it took a long time to get there, but we finally got there. So it's like in the Truman Show, he finally gets to the outer edge of the set of the set of the set of the set. So yeah. So what happened to our quantum entanglement? What's different about the world?

Yeah, so at the macroscopic level, I think probably not a huge amount, like the entanglement is only at a very small scale. However, there are a range of processes that we're starting to have hints at that entanglement may be really important. There's some biological processes such as people are postulating the stability of DNA maybe due to entanglement within the molecules itself. navigation of birds. Some birds use magnetic sensors and there's hints that there's quantum elements of that that rely on entanglement. Migration, migration of birds. Migration of birds, yeah. Navigation during migration is kind of what I mean. Like when they migrate, how do they know to go north? Yeah, how do they know which way to go? And so there's thoughts of that. And even things such as I think photosynthesis is the latest one that they're looking at that may have elements. Now, I think the jury is still out on all of these. Again, it's not quite my area of expertise, but I believe the jury is still out on all of these as to whether it's definitely quantum entanglement enhanced. But there's definitely some evidence that starting to point towards some of these biological processes, quantum physics is really important and entanglement in particular. And so, yeah, well, at first glance, you might think that if the programmer of the universe turned off entanglement, we wouldn't see any difference. It may actually be really important.

Okay, very cool. I love the idea of photosynthesis and quantum entanglement. That sounds so cool. All right, this is Bruce Lessie. And Bruce says, Dr. Tyson's and Hodgman, this is Bruce Lessie from Cripple

Creek, Virginia. My question is related to entanglement and spooky action at a distance as Einstein phrased it. In space time, for a photo traveling at the speed, he said photo, but I think he means photon. traveling at the speed of light represents zero time experienced by the photon. Because at the speed of the light, distance is non-existent. Why did Einstein have a problem with instantaneous communications between entangled particles? Nothing is at a distance for particles that travel at the speed of light. Yeah, so I'd say that's one of the reasons why it's really important to measure entanglement for things not just that aren't photons. So yeah, because photons travel at the speed of light, if you just measure entanglement with that, you could perhaps come up with some sort of explanation like that. But in our system, we measure it with atoms. Our atoms move very slowly. We give them a kick and they move at several centimeters per second. Let's call it an inch per second for you Americans with your freedom units.

Freedom units. Freedom units. That's exactly what we used to measure the octagon on the White House lawn. Freedom units. So yes, back to the atoms. So because they're moving relatively slowly, you can't come up with that sort of an explanation to explain it. and simultaneous, instantaneous communication suddenly becomes a problem again. Well, that's really cool, man, because, yeah, they have mass and, yeah, they're entangled at much, much, much slower speeds. So, yeah, that's really... Well, it's still a good question. That's a great question. Great question. Yeah, really good question, Bruce. Thanks. All right, this is David Barlow. And David Barlow says, greetings, Dr. Hosman, Dr. Tyson Lorde Nice. David Barlow from Chicago, Illinois here, a newly signed up Patreon supporter. Kudos to you, my friend. He says, I was wondering if, as an experimental physicist and an observer within the quantum field, you and your associates knew for a certain, for certain that you were not affecting the results of your experiments in the Bell's theorem. What loophole prevention precautions were taken to negate your field collapse of the wave function when taking measurements? Love your fantastic science broadcast guy. So yeah, how you know this that you're not messing it up like.

You always have a big doubt that, yeah, did we just do something wrong when measure something? That's why we have to do sort of rigorous and multiple tests. And for the scientific method in general, it's part of the

reason why you have to publish these papers that other scientists can then go, hang on, did you think about this? Did you try turning this off? Did you try turning that on? And yeah, it's an important part of the process. For our particular experiment, so maybe I should just briefly cover what loopholes are that are referred to there. So with Bell experiments, there's these things called loopholes, which are basically ways that people come up to say, well, maybe to still preserve locality, to still preserve the fact that you don't have instantaneous communication. And so it can be things such as, well, maybe Maybe the experiments conspiring in a way such that it only lets certain results through. Or maybe the observer is communicating in a way. If you set the interference on the true halves of the system and the measurement, if you don't set them in a way that they could communicate with each other, maybe there would be some really weird theory that could explain that.

And so all these loopholes have been closed with experiments on photons. For our particular experiment, we didn't close all these loopholes. So our experiment isn't loophole free. In principle, you could make some of these criticisms about our experiment. However, the fact that they've been closed with photons means that we probably expect that they should also be closed for atoms as well. And so, yeah. And part of the reason why we didn't close them was because a couple of them are technically extremely challenging to close with atoms, just because atoms move a lot slower than photons and they move over much smaller distances. And so, yeah, it's an ongoing area of work where it would be a good area of future work for groups like us. That's a damn good question. Chuck, we have time for a couple more. Cleo Fox. He says, hello, Dr. Tyson. My name is Cleo from Denver, Colorado. Modern quantum physics has achieved extraordinary predictive accuracy, but many of its foundational interpretations remain experimentally indistinguishable. Given recent advances in quantum information theory, weak measurements, quantum computing, and tests of non-locality, do you think the next major breakthrough in quantum physics is more likely to come from developing new mathematical frameworks or from entirely new experimental methodologies capable of probing quantum phenomena in ways we cannot.

currently access. Put another way. Are we currently limited by our theories or by the tools we use to test them? Yeah. So I was going to say something

similar to that in summary of that question. So Sean, there's, you know, philosophers like believing they have access to emerging truths in science in general, but especially in quantum physics, where there's so much that makes no freaking sense. So is there room for philosophers to guide the physicists through this and or out of it? Or we just stuck just as they say, shut up and calculate. And so that's a nuanced way of saying, was it was the final question there? Are we currently limited by our theories and the tools we use to test them?

on the shut up and calculate versus philosophy debate. I'm a big believer as an experimenter. If I'm a big fan of shut up and calculate, experiments are hard enough as is. Count me on that vote as well. Okay. Yeah. It's basically the maths tells us the results that give us predictions about experiments. And it works. Yeah. And it works. And so at that level, yeah, I'm happy with that. I'm happy to leave the questions to the essentially to the realm of philosophy to an extent. If you're If you can't actually make predictions of what an experiment will give you, I think that's basically philosophy. And I think there's definitely a role for that. I mean, we've seen that how brain bending some of these quantum effects are. And I think it's really interesting to probe that. I think it's really interesting to have philosophers in the light guide that and theorists and interact with quantum theorists. But I think it's also equally really important for experimentalists to actually test these results. And if you can, if you can't.

test these results, then you probably need to work harder on your theory, I think. I spent some time at Princeton where they're very theory-based, although they do quite a bit of experiments there. They have a tokamak and fusion reactor, but there's a strong theoretical legacy in the department. And there's a sign up somewhere or someone's door. It says, never trust an observation unless it's backed up by a good theory. So I think coming back to the question, it really is, it's both. There's a lot of work on theory, but there's also a lot of work on experiments to cover that. Do they inform one another, I think is kind of also the spirit of the question.

That's the correct way to think about that, Chuck. Absolutely. There's plenty of times, like our experimental entanglement, that was originally proposed by some theorists. We tried to do it the way they proposed and it didn't

work. So then we came up with a slightly different way and then we came back to them and then they helped analyze our results. And yeah, it's really, there's a lot of back and forth. Okay, cool. Go ahead and science. Cool. Chuck, one more question. We got time for it. All right, let's close it out with Melanie Stickler. And Melanie says, hello, Dr. Tyson. I'm Melanie, originally from Austria, now in the Bay Area of California. My question is about Dr. Hodgman's helium experiment. I understand massless photons have wave particle duality, but how does a massive particle like a helium atom, which is subject to gravity, function as a wave cloud? Furthermore, if measuring a quantum system collapses the wave function, how did the team measure the atoms in simultaneous momentum states without instantly destroying the superposition. I'm sure the paper covers this, but I'm having a hard time wrapping my head around it. So thank you. Yeah, helium is a massive particle compared to stuff we're used to. And so there's a wave function associated with such a massive particle.

Yeah, definitely. And it's a great question. And yeah, there's a wave function associated with helium atoms. And that's why we need to cool them down to make it work because otherwise at room temperature, the wave function is so small that you can't see it. But at these temperatures, the wave function is macroscopic. It's sort of in the order of tens to hundreds of micrometers. So that's a 0.1 of a millimeter. And yeah, it's quite large at that scale. The other part of the question, if I remember if I'm getting it correctly, was how do you measure that they're in two different states at once when they're only ever going to be in? How do you prove they're in two different states at once when you can only have a measure one result? And again, that really comes down to John Bell's work for how you can measure this Bell inequality where if you interfere those states, you can measure the results of that in the outcomes you get. You can interfere those states and you can get more probability of being in one than the other in your output due to the fact that you were in this superposition of two states at the same time. Is there a quantum entanglement arms race in the world? Like, who's leading the quantum entanglement experiments? Because I don't think it's us. It's us, Americans. You're in Australia. Who's ahead and who's behind? It kind of depends what you're talking about. I mean, our experiments probing the fundamentals of entanglement, there's there's still work going on in that. But a lot of what current quantum research is going into is how we do

something useful with entanglement. So how you can use it, so things such as quantum computing. So quantum computing is a computer that rather than using bits to encode information, so bits have to be one or zero, you use qubits, which can be in a superposition of one and zero at the same time. Then if you have multiples of these together, you can entangle the different qubits.

And by the fact that you can have qubits in many states at once, for certain types of problems, you can probe many answers at once, even though you only ever get one result when you get out, when you do your final measurement. So in principle, it seems really promising that you should be able to do much faster calculations, much higher level computation with this. But the problem is that finding And there's only particular problems that we know of that this is true. So there's a lot of work going into that. Then the other problem is that quantum systems are really hard to get to work on a large scale. So I have lots of qubits. And so there's a lot of work going into building these processes. And so yeah, there really is, if you want to call it an arms race, there's a lot of government and private investment in this at the moment. There's a lot of quantum computing startups all around the world. And the assumption that whoever gets advances in it first might have a leg up either economically or with regard to security or Computing and so even if at the end of the day It's just a pipe dream because it's like it's just a fun physics exercise But it doesn't have any practical use no one knows that yet. Is that a correct way to think about that?

Absolutely. It's still really an open question as to what the impact of quantum computing will be. Quantum computing could be anything from, like you say, massive economic implications, massive security implications. It could be used to things such as medical implications like drug development, data processing, all these massive things. Or it could just be on a much smaller level and that it's kind of a toy physics system that helps us advance physics, but may not have quite such a wide economic implication. And I think it's a really, really exciting time. I bet people said the same thing about quantum physics a century ago. This is just a curiosity on the fringes of physics. We'll never have any use for this. But it's still fun anyway. And now it's the foundation of our IT revolution. Everything.

everything. Yeah. It's one of the reasons it's really important to invest in basic research. I mean, the work my group does is we don't know what that is anymore. In the United States, we don't know what that is anymore. Exactly. We don't do research here anymore. Okay. Australia's not much better, unfortunately. We run on vibes. Vibes. No, right now it's on physics fumes. That's all that's left. Yeah, unfortunately. Well, sure, that's all the time we have. I'm saddened by this because this topic has no end of curious people out there thinking about it. They've read about it, and it's not every day you get to bump into someone who gets paid for thinking about it. Congratulations on your Bose-Einstein condensate as a source of entanglement. Okay, we will look for our invitation to Stockholm in the mail. Really good, and thanks a lot for the questions. And yeah, thanks for having me on your show. I am Neil deGrasse Tyson, your personal astrophysicist, finishing up a very special edition of Cosmic Queries, specializing in quantum entanglement.

Until next time, keep looking up.